

DBMS

Why This Matters Now

- DBMS is high-value for HPCL and directly transferable to IFFCO.
- SQL was already exposed as a weakness in BARC, so this is a high-return revision area.

High-Yield Topics

- ER model and relational model
- Relational algebra
- Normalization up to BCNF
- SQL joins, subqueries, grouping, HAVING, NULL traps
- Transactions, ACID, serializability
- Concurrency control and locking
- Indexing, file organization, backup and recovery

IFFCO Carry-Over

- Shared base: SQL logic, transactions, indexing
- Extra gap for IFFCO: Oracle syntax habits, PL/SQL stored objects, DBA awareness

Common Traps

- Partial vs transitive dependency confusion
- COUNT(*) vs COUNT(column)
- WHERE vs HAVING misuse
- Weak conflict-serializability graph handling

Done For Now Means

- one normalization rules sheet
- one SQL traps sheet
- 15 SQL/output or transaction questions